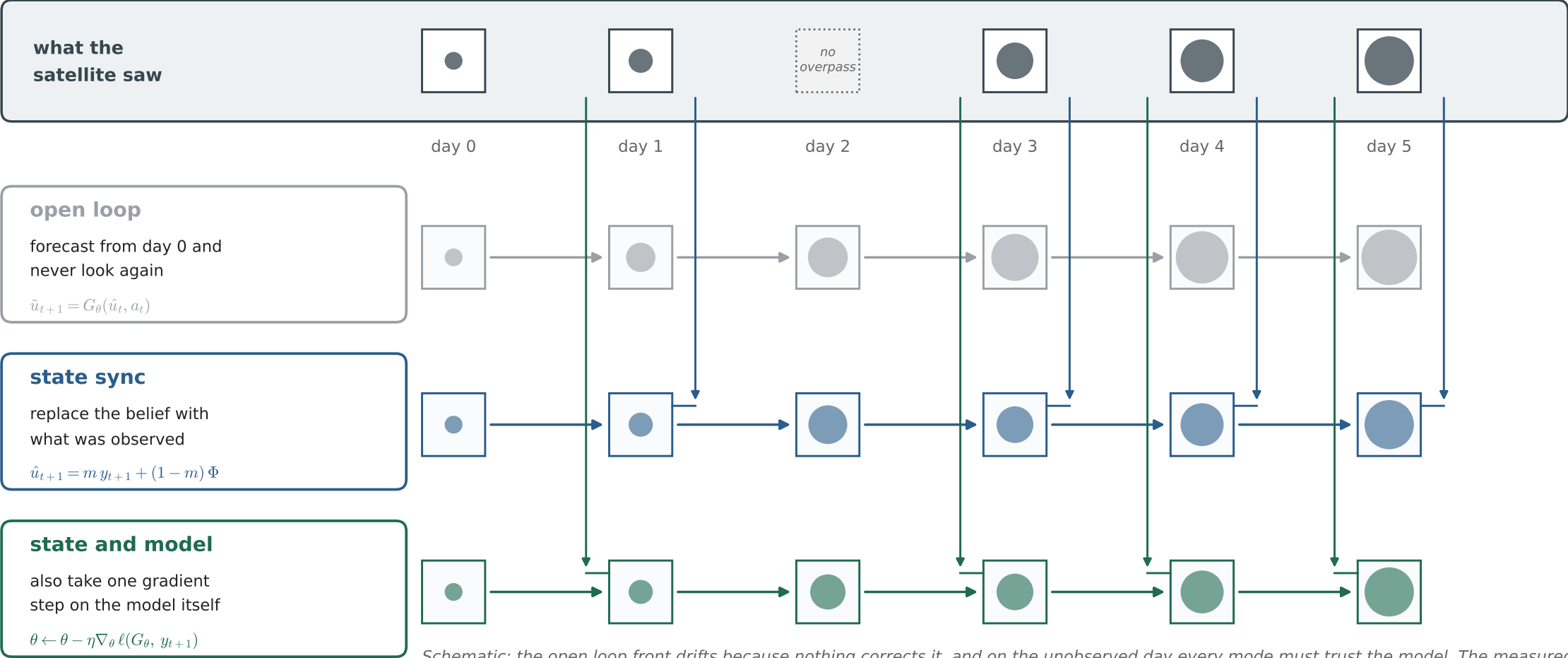


Three ways to carry a state across real satellite days

Eight large wildfires, VIIRS detections, leave one fire out, 3 seeds. Scored as average precision against the detections actually recorded.



Schematic: the open loop front drifts because nothing corrects it, and on the unobserved day every mode must trust the model. The measured skill is tabulated below.

Average precision by lead time, and the paired test across the eight fires

	day +1	day +2	day +3	
open loop	0.073	0.047	0.038	
state sync	0.608	0.237	0.085	gain of 0.535 at day 1, better on 8 of 8 fires, t of 15.9
state and model	0.613	0.288	0.110	adds 0.051 at day 2 and 0.025 at day 3, better on 8 of 8 fires